Ransomware protection on Windows with Wazuh

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Ransomware

Ransomware is a type of malicious software designed to encrypt or lock files on a computer or network, making them inaccessible to the user until a ransom is paid. It is one of the most critical cybersecurity threats today because it can affect both personal users and organizations. The primary objective of ransomware is financial gain, but its impact goes far beyond financial loss.

The key consequences of ransomware include:

Data Loss: Files such as documents, images, and databases become unreadable and cannot be restored without the attacker's decryption key.
Operational Disruption: Systems and applications may be rendered unusable, causing downtime and interruptions in workflows.
Financial Impact: In addition to ransom payments, organizations face costs related to system recovery, data restoration, and lost productivity.
Reputation Damage: Exposure of sensitive data or prolonged system unavailability can erode trust among customers, partners, and stakeholders.
Network Propagation: Ransomware often spreads laterally across connected systems, increasing the scale of damage.

Ransomware attacks are fast, unpredictable, and often difficult to reverse, making early detection and containment critical for minimizing damage.

Role of Wazuh in Ransomware Mitigation

Wazuh is a powerful open-source security monitoring platform that provides the ability to detect, analyze, and respond to threats. In the context of ransomware, Wazuh can be configured to continuously monitor critical files and system logs, giving security teams the ability to respond before significant damage occurs.

The main objectives of implementing Wazuh for ransomware protection are:

Real-Time File Monitoring:
On Windows Server, critical directories such as:
C:\Users\ (user profiles and personal files)

C:\Windows\System32\ (system executables and configuration files)
C:\Program Files\ and C:\Program Files (x86)\ (application directories)
C:\inetpub\wwwroot\ (if the server hosts web applications)
These are monitored for changes such as creation, modification, deletion, or renaming of files. Suspicious activities like sudden mass renaming or creation of encrypted files trigger immediate alerts.

Noise Reduction:
Wazuh can ignore non-critical locations such as C:\Windows\Temp\ or browser cache directories, ensuring alerts remain meaningful and reduce false alarms.
Scheduled Full Scans:
In addition to real-time monitoring, Wazuh performs scheduled scans (for example, every 12 hours) to catch anything missed during live monitoring.
Log Monitoring and Analysis:
On Windows Server, important logs include:
Windows Security Event Logs → Failed logins, privilege escalation attempts, suspicious user activity.
System Logs (Event Viewer) → Service failures, abnormal shutdowns, or ransomware-related errors.
Application Logs → Issues in running applications that could hint at ransomware behavior.

This combination of File Integrity Monitoring (FIM) and log analysis ensures ransomware activity can be detected early, understood in context, and acted upon quickly.

Achievements and Benefits

Implementing this Wazuh configuration provides several significant benefits:

Early Detection: Suspicious file changes are detected immediately, allowing rapid intervention before the attack escalates.
Improved Visibility: Monitoring both files and logs gives security teams a full view of ransomware activity.
Event Correlation: Alerts from file changes and logs can be correlated to detect patterns and the source of malicious activity.
Reduced False Positives: By focusing on important directories and logs, alerts remain meaningful and actionable.
Foundation for Automated Response: Prepares the system for Active Responses

such as quarantining suspicious files, blocking malicious IPs, or isolating compromised Windows hosts.

Objective

Build a full ransomware detection and mitigation system with Wazuh that:

Monitors for ransomware behavior (using file integrity monitoring).
Detects suspicious extensions and mass file changes indicating encryption.
Automatically quarantines suspicious files (not deleting to avoid data loss).
Automatically blocks attacker IPs on related network attacks (e.g., RDP brute force or failed login attempts).

Setup Environment

Wazuh Manager: Kali Linux machine
Wazuh Agent: Windows Server machine

Pre-requisites

Wazuh manager and Windows agent installed and communicating.
Ensure firewall ports are open:

UDP 1514 → Event transmission (agent → manager)
TCP 1515 → Agent registration

Check services and connectivity:
On **Manager (Kali)**, ensure the agent is successfully registered and actively sending logs/events.

```
sudo systemctl status wazuh-manager
```

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl status wazuh-manager

● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
     Active: active (running) since Thu 2025-09-04 16:44:44 PKT; 57min ago
 Invocation: d6ffbb12537c4ffbb9653ad5425cb657
    Process: 1300 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
      Tasks: 178 (limit: 5614)
     Memory: 355.7M (peak: 1.4G, swap: 160.7M, swap peak: 161M)
        CPU: 1min 53.655s
     CGroup: /system.slice/wazuh-manager.service
```

Windows endpoint

On the Windows endpoint, we configure the Wazuh FIM module and a Wazuh Active Response script that removes malicious files.

File integrity monitoring configuration

Perform the following steps to configure the Wazuh FIM module to monitor near real-time changes in the folders.

Run this command to open the configuration file in Notepad:

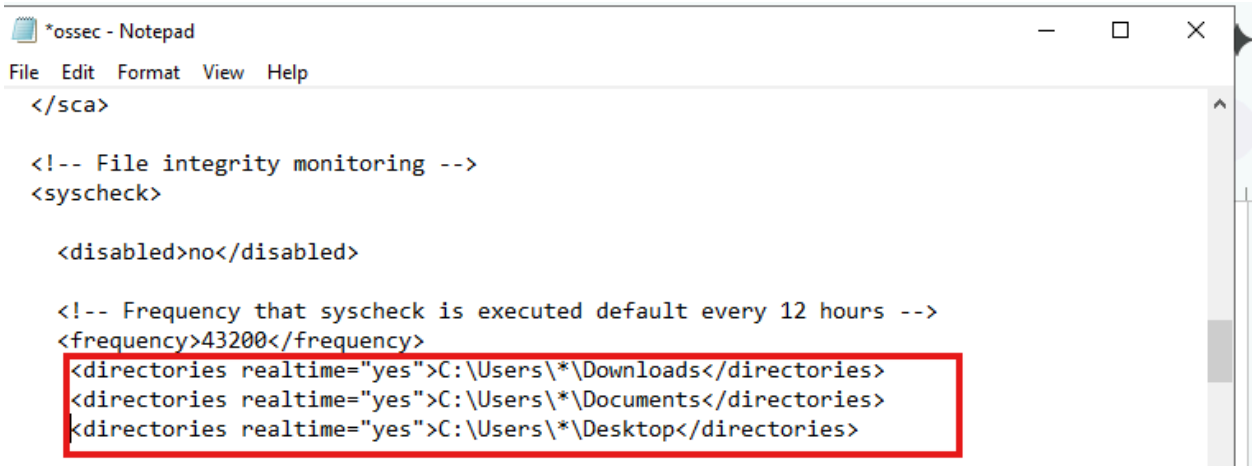
notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"

Add the below entry within the <syscheck> block to monitor the Downloads, Documents, and Desktop folders of all users for changes:

<directories realtime="yes">C:\Users*\Downloads</directories>

<directories realtime="yes">C:\Users*\Documents</directories>

<directories realtime="yes">C:\Users*\Desktop</directories>



Malware removal Wazuh Active Response script

Perform the following steps to create a Wazuh Active Response script that removes files detected as malicious by VirusTotal.

This command will **download the installer file** from Python's official website into C:\:

Invoke-WebRequest -Uri "https://www.python.org/ftp/python/3.12.5/python-3.12.5-amd64.exe" -OutFile "C:\python-installer.exe"

```
PS C:\Users\administrator> Invoke-WebRequest -Uri "https://www.python.org/ftp/python/3.12.5/python-3.12.5-amd64.exe" -Ou
tFile "C:\python-installer.exe"
PS C:\Users\administrator>
```

This will run the installer without opening the GUI and will automatically check both boxes for you:

Start-Process "C:\python-installer.exe" -ArgumentList "/quiet InstallAllUsers=1 PrependPath=1" -Wait

```
PS C:\Users\administrator> Start-Process "C:\python-installer.exe" -ArgumentList "/quiet InstallAllUsers=1 PrependPath=1
" -Wait
PS C:\Users\administrator>
```

Check if Python was installed successfully:

py --version

```
PS C:\Users\administrator> py --version
Python 3.12.5
PS C:\Users\administrator>
```

Create a Wazuh Active Response script remove-threat.py in the:

C:\Program Files (x86)\ossec-agent\active-response\bin\

directory. This script removes any file detected as malicious by the VirusTotal integration:

So go into the following directory and create file.

cd "C:\Program Files (x86)\ossec-agent\active-response\bin"

Create file:

New-Item -Path "remove-threat.py" -ItemType File

```
PS C:\Users\administrator> cd "C:\Program Files (x86)\ossec-agent\active-response\bin"
PS C:\Program Files (x86)\ossec-agent\active-response\bin> New-Item -Path "remove-threat.py" -ItemType File


    Directory: C:\Program Files (x86)\ossec-agent\active-response\bin


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          9/5/2025   4:35 PM              0 remove-threat.py


PS C:\Program Files (x86)\ossec-agent\active-response\bin>
```

And paste code in given file by following command:

notepad.exe "remove-threat.py"

```
PS C:\Program Files (x86)\ossec-agent\active-response\bin> notepad.exe "remove-threat.py"
PS C:\Program Files (x86)\ossec-agent\active-response\bin>
```

#!/usr/bin/python3

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All rights reserved.

```
import os

import sys

import json

import datetime


if os.name == 'nt':

    LOG_FILE = "C:\\Program Files
(x86)\\ossec-agent\\active-response\\active-responses.log"

else:

    LOG_FILE = "/var/ossec/logs/active-responses.log"


ADD_COMMAND = 0

DELETE_COMMAND = 1

CONTINUE_COMMAND = 2

ABORT_COMMAND = 3


OS_SUCCESS = 0

OS_INVALID = -1


class message:

    def __init__(self):

        self.alert = ""

        self.command = 0
```

```
def write_debug_file(ar_name, msg):
    with open(LOG_FILE, mode="a") as log_file:
        log_file.write(str(datetime.datetime.now().strftime('%Y/%m/%d %H:%M:%S')) + " " +
ar_name + ": " + msg +"\n")


def setup_and_check_message(argv):


    # get alert from stdin
    input_str = ""
    for line in sys.stdin:
        input_str = line
        break



    try:
        data = json.loads(input_str)
    except ValueError:
        write_debug_file(argv[0], 'Decoding JSON has failed, invalid input format')
        message.command = OS_INVALID
        return message

    message.alert = data
```

```
    command = data.get("command")


    if command == "add":
        message.command = ADD_COMMAND
    elif command == "delete":
        message.command = DELETE_COMMAND
    else:
        message.command = OS_INVALID
        write_debug_file(argv[0], 'Not valid command: ' + command)


    return message



def send_keys_and_check_message(argv, keys):


    # build and send message with keys
    keys_msg = json.dumps({"version": 1,"origin":{"name":
argv[0],"module":"active-response"},"command":"check_keys","parameters":{"keys":keys}})


    write_debug_file(argv[0], keys_msg)


    print(keys_msg)
    sys.stdout.flush()
```

```
# read the response of previous message
input_str = ""
while True:
    line = sys.stdin.readline()
    if line:
        input_str = line
        break

# write_debug_file(argv[0], input_str)

try:
    data = json.loads(input_str)
except ValueError:
    write_debug_file(argv[0], 'Decoding JSON has failed, invalid input format')
    return message

action = data.get("command")

if "continue" == action:
    ret = CONTINUE_COMMAND
elif "abort" == action:
    ret = ABORT_COMMAND
else:
    ret = OS_INVALID
```

```
        write_debug_file(argv[0], "Invalid value of 'command'")

    return ret

def main(argv):

    write_debug_file(argv[0], "Started")

    # validate json and get command
    msg = setup_and_check_message(argv)

    if msg.command < 0:
        sys.exit(OS_INVALID)

    if msg.command == ADD_COMMAND:
        alert = msg.alert["parameters"]["alert"]
        keys = [alert["rule"]["id"]]
        action = send_keys_and_check_message(argv, keys)

        # if necessary, abort execution
        if action != CONTINUE_COMMAND:

            if action == ABORT_COMMAND:
                write_debug_file(argv[0], "Aborted")
```

```
            sys.exit(OS_SUCCESS)
        else:
            write_debug_file(argv[0], "Invalid command")
            sys.exit(OS_INVALID)

    try:
        file_path = msg.alert["parameters"]["alert"]["data"]["virustotal"]["source"]["file"]
        if os.path.exists(file_path):
            os.remove(file_path)
        write_debug_file(argv[0], json.dumps(msg.alert) + " Successfully removed threat")
    except OSError as error:
        write_debug_file(argv[0], json.dumps(msg.alert) + "Error removing threat")



  else:
    write_debug_file(argv[0], "Invalid command")

  write_debug_file(argv[0], "Ended")

  sys.exit(OS_SUCCESS)

if __name__ == "__main__":
  main(sys.argv)
```

*remove-threat - Notepad

File Edit Format View Help

```
#!/usr/bin/python3
# Copyright (C) 2015-2022, Wazuh Inc.
# All rights reserved.

import os
import sys
import json
import datetime

if os.name == 'nt':
    LOG_FILE = "C:\\Program Files (x86)\\ossec-agent\\active-response\\active-responses.log
else:
    LOG_FILE = "/var/ossec/logs/active-responses.log"

ADD_COMMAND = 0
DELETE_COMMAND = 1
CONTINUE_COMMAND = 2
ABORT_COMMAND = 3

OS_SUCCESS = 0
OS_INVALID = -1

class message:
    def __init__(self):
```

Run the following commands to install pyinstaller and convert the remove-threat.py Wazuh Active Response script to a Windows executable:

& "C:\Program Files\Python312\python.exe" -m pip install --upgrade pip

& "C:\Program Files\Python312\python.exe" -m pip install pyinstaller

```
PS C:\Program Files (x86)\ossec-agent\active-response\bin> & "C:\Program Files\Python312\python.exe" -m pip install --upg
rade pip
Requirement already satisfied: pip in c:\program files\python312\lib\site-packages (24.2)
Collecting pip
  Downloading pip-25.2-py3-none-any.whl.metadata (4.7 kB)
Downloading pip-25.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 13.7 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.2
    Uninstalling pip-24.2:
      Successfully uninstalled pip-24.2
```

& "C:\Program Files\Python312\python.exe" -m PyInstaller -F "remove-threat.py"

```
PS C:\Program Files (x86)\ossec-agent\active-response\bin> & "C:\Program Files\Python312\python.exe" -m PyInstaller -F "r
emove-threat.py"
285 INFO: PyInstaller: 6.15.0, contrib hooks: 2025.8
285 INFO: Python: 3.12.5
300 INFO: Platform: Windows-2022Server-10.0.20348-SP0
300 INFO: Python environment: C:\Program Files\Python312
300 INFO: wrote C:\Program Files (x86)\ossec-agent\active-response\bin\remove-threat.spec
316 INFO: Module search paths (PYTHONPATH):
['C:\\Program Files (x86)\\ossec-agent\\active-response\\bin',
 'C:\\Program Files\\Python312\\python312.zip',
 'C:\\Program Files\\Python312\\DLLs',
 'C:\\Program Files\\Python312\\Lib',
 'C:\\Program Files\\Python312',
 'C:\\Program Files\\Python312\\Lib\\site-packages',
 'C:\\Program Files\\Python312\\Lib\\site-packages\\setuptools\\_vendor',
 'C:\\Program Files (x86)\\ossec-agent\\active-response\\bin']
662 INFO: checking Analysis
675 INFO: Building Analysis because Analysis-00.toc is non existent
675 INFO: Running Analysis Analysis-00.toc
675 INFO: Target bytecode optimization level: 0
675 INFO: Initializing module dependency graph...
676 INFO: Initializing module graph hook caches...
706 INFO: Analyzing modules for base_library.zip ...
1849 INFO: Processing standard module hook 'hook-heapq.py' from 'C:\\Program Files\\Python312\\Lib\\site-packages\\PyInst
aller\\hooks'
```

The second command creates a remove-threat.exe executable in a folder called **dist** in your working directory.

```
PS C:\Program Files (x86)\ossec-agent\active-response\bin> cd dist
PS C:\Program Files (x86)\ossec-agent\active-response\bin\dist> dir


    Directory: C:\Program Files (x86)\ossec-agent\active-response\bin\dist


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----          9/5/2025   4:47 PM        7281054 remove-threat.exe


PS C:\Program Files (x86)\ossec-agent\active-response\bin\dist>
```

Copy the remove-threat.exe executable file to the:

C:\Program Files (x86)\ossec-agent\active-response\bin\

Directory.

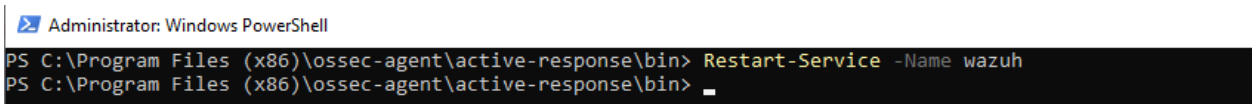
Command:

Copy-Item ".\dist\remove-threat.exe" "C:\Program Files (x86)\ossec-agent\active-response\bin\" -Force

```
Administrator: Windows PowerShell
PS C:\Program Files (x86)\ossec-agent\active-response\bin> Copy-Item ".\dist\remove-threat.exe" "C:\Program Files (x86)\o
ssec-agent\active-response\bin\" -Force
PS C:\Program Files (x86)\ossec-agent\active-response\bin>
```

Restart the Wazuh agent using PowerShell with administrative privilege to apply the configuration changes:

```
Restart-Service -Name wazuh
```



Wazuh server

We configure the Wazuh VirusTotal integration to scan modified or added files. We also configure the Wazuh Active Response module to trigger the remove-threat.exe executable.

Wazuh VirusTotal integration

Perform the following step on the Wazuh server to configure the VirusTotal integration.

Add the following configuration within the <ossec> block of the /var/ossec/etc/ossec.conf file to enable the VirusTotal integration. Replace <YOUR_VIRUS_TOTAL_API_KEY> with your VirusTotal API key. This allows Wazuh to trigger a VirusTotal query on the file whenever a file addition or modification is detected:

```
sudo nano /var/ossec/etc/ossec.conf
```

```
<integration>
  <name>virustotal</name>
  <api_key><YOUR_VIRUS_TOTAL_API_KEY></api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
```

Note: The free VirusTotal API rate limits requests to four per minute. If you have a premium VirusTotal API key, with a high frequency of queries allowed, you can configure Wazuh to monitor more directories besides the Downloads, Desktop and Documents.

Wazuh Active Response module configuration

Perform the following steps on the Wazuh server to configure the Wazuh Active Response module.

Add the following configuration within the <ossec> block of the /var/ossec/etc/ossec.conf file to configure the Wazuh Active Response command:

```
<command>
  <name>remove-threat</name>
  <executable>remove-threat.exe</executable>
  <timeout_allowed>no</timeout_allowed>
</command>


<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
```

When VirusTotal detects malware written to disk, remove-threat.exe is triggered to remove the malware from the monitored endpoint.

```
GNU nano 8.3                                        /var/ossec/etc/ossec.conf *
<command>
  <name>remove-threat</name>
  <executable>remove-threat.exe</executable>
  <timeout_allowed>no</timeout_allowed>
</command>

<active-response>
  <disabled>no</disabled>
  <command>remove-threat</command>
  <location>local</location>
  <rules_id>87105</rules_id>
</active-response>
```

Custom rules configuration

Perform the following steps to create custom rules that trigger alerts when the Wazuh Active Response script successfully or unsuccessfully removes a threat file.

Create a virustotal_rules.xml rule file in the /var/ossec/etc/rules/ directory:

touch /var/ossec/etc/rules/virustotal_rules.xml

```
┌──(kali㉿kali)-[~]
└─$ sudo touch /var/ossec/etc/rules/virustotal_rules.xml

┌──(kali㉿kali)-[~]
└─$ sudo nano /var/ossec/etc/rules/virustotal_rules.xml
```

Add the following custom rules to the /var/ossec/etc/rules/virustotal_rules.xml rule file:
sudo nano /var/ossec/etc/rules/virustotal_rules.xml

```
<group name="virustotal,">

  <rule id="100092" level="12">

      <if_sid>657</if_sid>

      <match>Successfully removed threat</match>

      <description>$(parameters.program) removed threat located at
$(parameters.alert.data.virustotal.source.file)</description>
```

```
  </rule>


  <rule id="100093" level="12">

    <if_sid>657</if_sid>

    <match>Error removing threat</match>

    <description>Error removing threat located at
$(parameters.alert.data.virustotal.source.file)</description>

  </rule>

</group>
```

```
  GNU nano 8.3                                        /var/ossec/etc/rules/virustotal_rules.xml
<group name="virustotal,">
  <rule id="100092" level="12">
      <if_sid>657</if_sid>
      <match>Successfully removed threat</match>
      <description>$(parameters.program) removed threat located at $(parameters.alert.data.virustotal.source.file)</description>
  </rule>

  <rule id="100093" level="12">
    <if_sid>657</if_sid>
    <match>Error removing threat</match>
    <description>Error removing threat located at $(parameters.alert.data.virustotal.source.file)</description>
  </rule>
</group>
```

Restart the Wazuh manager to apply the configuration changes:

```
sudo systemctl restart wazuh-manager
```

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart wazuh-manager


┌──(kali㉿kali)-[~]
└─$
```

Ransomware Execution Detection

Wazuh can detect some ransomware variants during its execution using custom rules. These rules monitor for common activities associated with ransomware attacks such as deleting volume shadow copies or restricting system recovery.

Windows Endpoint

Perform the following steps to install and configure Sysmon. Sysmon is a Windows system service that logs detailed information on system activities like process creation, network connections, and file access for security monitoring.

Sysmon Configuration

Run the following commands on PowerShell with admin privileges to install and configure Sysmon:

```
# Set Variables

$url = "https://download.sysinternals.com/files/Sysmon.zip"

$outputPath = "C:\Temp\Sysmon.zip"

$extractPath = "C:\Temp"

$configUrl =
"https://wazuh.com/resources/blog/emulation-of-attack-techniques-and-detection-with-wazuh/sysmonconfig.xml"

$configPath = "$extractPath\sysmonconfig.xml"


# Install Sysmon

New-Item -ItemType Directory -Path C:\Temp

Invoke-WebRequest -Uri $url -OutFile $outputPath

Expand-Archive -Path $outputPath -DestinationPath $extractPath

wget -Uri $configUrl -OutFile $configPath

& "$extractPath\Sysmon64.exe" -accepteula -i $configPath
```

```
Administrator: Windows PowerShell
PS C:\Users\administrator> # Set Variables
PS C:\Users\administrator> $url = "https://download.sysinternals.com/files/Sysmon.zip"
PS C:\Users\administrator> $outputPath = "C:\Temp\Sysmon.zip"
PS C:\Users\administrator> $extractPath = "C:\Temp"
PS C:\Users\administrator> $configUrl = "https://wazuh.com/resources/blog/emulation-of-attack-techniques-and-detection-w
ith-wazuh/sysmonconfig.xml"
PS C:\Users\administrator> $configPath = "$extractPath\sysmonconfig.xml"
PS C:\Users\administrator>
PS C:\Users\administrator> # Install Sysmon
PS C:\Users\administrator> New-Item -ItemType Directory -Path C:\Temp


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          9/5/2025   4:58 PM                Temp


PS C:\Users\administrator> Invoke-WebRequest -Uri $url -OutFile $outputPath
PS C:\Users\administrator> Expand-Archive -Path $outputPath -DestinationPath $extractPath
PS C:\Users\administrator> wget -Uri $configUrl -OutFile $configPath
PS C:\Users\administrator> & "$extractPath\Sysmon64.exe" -accepteula -i $configPath


System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
```

Add the following configuration within the <ossec_config> block of the C:\Program Files (x86)\ossec-agent\ossec.conf file to collect Sysmon logs and forward them to the Wazuh server:

notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"

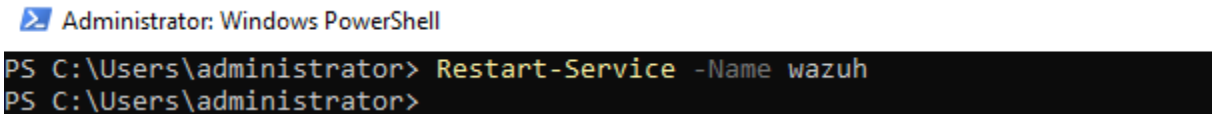
```
Administrator: Windows PowerShell
PS C:\Users\administrator> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
```

```
<localfile>

  <location>Microsoft-Windows-Sysmon/Operational</location>

  <log_format>eventchannel</log_format>

</localfile>
```

```
*ossec - Notepad
File  Edit  Format  View  Help
  <localfile>
      <location>Microsoft-Windows-Sysmon/Operational</location>
      <log_format>eventchannel</log_format>
  </localfile>
```

Restart the Wazuh agent using **PowerShell with administrative privilege** to apply the configuration changes:

Restart-Service -Name wazuh



Wazuh Server

On the Wazuh server, configure **custom rules** to generate alerts when ransomware execution activities are detected.

Custom Rules Configuration

Create a ransomware_rules.xml rule file in the /var/ossec/etc/rules/ directory:
sudo touch /var/ossec/etc/rules/ransomware_rules.xml

Add the following custom rules to the /var/ossec/etc/rules/ransomware_rules.xml rule file. These rules are made up of a group of likely ransomware actions observed on affected endpoints and are not specific to one type of malware. The rules may trigger or not trigger depending on the ransomware:

sudo nano /var/ossec/etc/rules/ransomware_rules.xml

<group name="malware,ransomware,ransomware_pre_detection">

<!-- Rules to detect Ransomware attack -->

```
<!-- Suspicious command execution -->

<rule id="100600" level="12">

  <if_sid>61603</if_sid>

  <field name="win.eventdata.parentCommandLine"
type="pcre2">(?i)[c-z]:\\\\Windows\\\\System32\\\\svchost\.exe\s-k\sWerSvcGroup</fi
eld>

  <field name="win.eventdata.commandLine"
type="pcre2">(?i)[c-z]:\\\\Windows\\\\system32\\\\WerFault\.exe\s-pss\s-s\s\d+\s-p\s\
d+\s-ip\s\d+</field>

  <description>Possible WerFault DLL Sideloading
$(win.eventdata.commandLine).</description>

  <mitre>

    <id>T1546.008</id>

  </mitre>

</rule>


<rule id="100601" level="10" >

  <if_sid>61603</if_sid>

  <field name="win.eventdata.parent.image.path"
type="pcre2">(?i)regedit.exe</field>

  <field name="win.eventdata.commandLine" type="pcre2">(?i)schtasks.exe
\/create.*\\cmd.exe.*start wordpad.exe.*.dll</field>

  <description>Suspicious scheduled task created.</description>

  <mitre>

    <id>T1546.008</id>
```

```
    </mitre>
  </rule>

  <rule id="100602" level="7">
    <if_sid>92027</if_sid>
    <field name="win.eventdata.CommandLine"
type="pcre2">Install-WindowsFeature.*RSAT-ADPowerShell</field>
    <description>Remote Server Administration Tools installed.</description>
    <mitre>
      <id>T1562</id>
    </mitre>
  </rule>

<!-- Impair defenses -->
  <rule id="100603" level="10">
    <if_sid>92042</if_sid>
    <field name="win.eventdata.CommandLine" type="pcre2">netsh advfirewall set
currentprofile state off</field>
    <description>Windows firewall disabled.</description>
    <mitre>
      <id>T1562</id>
    </mitre>
  </rule>
```

```
  <rule id="100604" level="10">

    <if_sid>61614</if_sid>

    <field name="win.eventdata.targetObject" type="pcre2"
>HKLM\\\\System\\\\CurrentControlSet\\\\Services\\\\WinDefend</field>

    <field name="win.eventdata.eventType" type="pcre2">^DeleteKey$</field>

    <field name="win.eventdata.user" type="pcre2" >NT
AUTHORITY\\\\SYSTEM</field>

    <description>Windows defender service $(win.eventdata.user) has been deleted
on $(win.system.computer). Possible malicious activity.</description>

    <mitre>

      <id>T1562.001</id>

    </mitre>

  </rule>



  <rule id="100605" level="10">

    <if_sid>92027,92021</if_sid>

    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)powershell.*New-ItemProperty.*Windows
Defender.*DisableAntiSpyware.*-Value 1.*</field>

    <description>Windows defender service has been deleted on
$(win.system.computer). Possible malicious activity.</description>

    <mitre>

      <id>T1562.001</id>

    </mitre>

  </rule>
```

```
  <rule id="100606" level="10">

    <if_sid>92008,92027</if_sid>

    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)powershell.*Set-MpPreference.*-DisableRealTimeMonitoring.*true
</field>

    <description>Windows defender realtime protection has been disabled on
$(win.system.computer). Possible malicious activity.</description>

    <mitre>

      <id>T1562.001</id>

    </mitre>

  </rule>



  <rule id="100607" level="10">

    <if_sid>92042</if_sid>

    <field name="win.eventdata.CommandLine" type="pcre2">reg.exe .*Windows
Defender\\Real-Time Protection.*Disable|\/d</field>

    <description>Windows defender realtime protection has been disabled on
$(win.system.computer). Possible malicious activity.</description>

    <mitre>

      <id>T1562</id>

    </mitre>

  </rule>



  <rule id="100608" level="10">
```

```
    <if_sid>92042</if_sid>

    <field name="win.eventdata.ruleName" type="pcre2">(?i)Disabling Security
Tools</field>

    <field name="win.eventdata.targetObject" type="pcre2">(?i)Windows
Defender</field>

    <description>Windows Defender feature disabled on $(win.system.computer).
Possible malicious activity</description>

    <mitre>

      <id>T1562</id>

    </mitre>

  </rule>



  <rule id="100609" level="10">

    <if_sid>92042</if_sid>

    <field name="win.eventdata.CommandLine" type="pcre2">dism .*
\/Disable-feature \/FeatureName:Windows-Defender</field>

    <description>Windows Defender disabled.</description>

    <mitre>

      <id>T1562</id>

    </mitre>

  </rule>



  <rule id="100610" level="10">

    <field name="win.system.providerName" type="pcre2">(?i)SecurityCenter</field>
```

```
    <field name="win.eventdata.data" type="pcre2">(?i)Windows Defender,
SECURITY_PRODUCT_STATE_SNOOZED</field>

    <description>Windows Defender snoozed on $(win.system.computer). Possible
malicious activity</description>

    <mitre>

      <id>T1562</id>

    </mitre>

  </rule>


<!-- System recovery inhibition -->

  <rule id="100611" level="10">

    <if_sid>61603</if_sid>

    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)bcdedit\s\s\/set\s{default}\sbootstatuspolicy\signoreallfailures</field
>

    <description>Boot configuration data edited.</description>

    <mitre>

      <id>T1059</id>

    </mitre>

  </rule>




<!-- Persistence detection -->

  <rule id="100612" level="10">
```

```
    <if_sid>92300</if_sid>

    <field name="win.eventdata.image" type="pcre2">(?i)\.exe</field>

    <field name="win.eventdata.eventType" type="pcre2">(?i)SetValue</field>

    <field name="win.eventdata.targetObject"
type="pcre2">(?i)HKLM\\\\SOFTWARE\\\\Microsoft\\\\Windows\\\\CurrentVersion\\\\R
un\\\\[A-Za-z0-9]+</field>

    <description>New run key added to registry by
$(win.eventdata.image).</description>

    <mitre>

      <id>T1547.001</id>

    </mitre>

  </rule>




 <rule id="100613" level="10">

    <if_sid>61613</if_sid>

    <field name="win.eventdata.image" type="pcre2">\.exe</field>

    <field name="win.eventdata.targetFilename"
type="pcre2">(?i)ProgramData\\\\Microsoft\\\\Windows\\\\Start
Menu\\\\Programs\\\\Startup\\\\.+\.exe</field>

    <description>$(win.eventdata.targetFilename) added to Startup programs by
$(win.eventdata.image).</description>

    <mitre>

      <id>T1547.001</id>

    </mitre>
```

```
  </rule>


  <rule id="100614" level="10">

    <field name="win.eventdata.ruleName" type="pcre2">(?i)Credential
Dumping</field>

    <field name="win.eventdata.sourceImage" type="pcre2">WerFault.exe</field>

    <description>WerFault abused to dump credentials.</description>

    <mitre>

      <id>T1003</id>

    </mitre>

  </rule>


<!-- System recovery inhibition -->

  <rule id="100615" level="12">

    <if_sid>61603</if_sid>

    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)vssadmin\s\sdelete\sshadows\s\/all\s\/quiet</field>

    <description>Volume shadow copy deleted using
$(win.eventdata.originalFileName). Potential ransomware activity
detected.</description>

    <mitre>

      <id>T1490</id>

      <id>T1059.003</id>

    </mitre>
```

```
  </rule>


  <rule id="100616" level="12">

    <if_sid>92032</if_sid>

    <field name="win.eventdata.parentCommandLine"
type="pcre2">(?i)vssadmin.*delete.*shadow</field>

    <description>Volume shadow copy deleted using
$(win.eventdata.originalFileName). Potential ransomware activity
detected.</description>

    <mitre>

      <id>T1490</id>

      <id>T1059.003</id>

    </mitre>

  </rule>


  <rule id="100617" level="12">

    <if_sid>61603</if_sid>

    <field name="win.eventdata.CommandLine" type="pcre2">(?i).*Shadowcopy
.*Delete</field>

    <description>Volume shadow copy deleted using
$(win.eventdata.originalFileName). Potential ransomware activity
detected.</description>

    <mitre>

      <id>T1490</id>

      <id>T1059.003</id>
```

```
    </mitre>
  </rule>


  <rule id="100618" level="12">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.CommandLine" type="pcre2">wmic shadowcopy
delete</field>
    <description>$(win.eventdata.originalFileName) invoked to delete shadow copies.
Potential ransomware activity detected.</description>
    <mitre>
      <id>T1490</id>
      <id>T1059.003</id>
    </mitre>
  </rule>


  <rule id="100619" level="12">
    <field name="win.system.providerName"
type="pcre2">(?i)Microsoft-Windows-Sysmon</field>
    <field name="win.eventdata.CommandLine" type="pcre2">(?i)delete
shadows</field>
    <description>Volume Shadow copy deleted on $(win.system.computer). Potential
ransomware activity detected.</description>
    <mitre>
      <id>T1490</id>
      <id>T1059.003</id>
```

```
    </mitre>
  </rule>

  <rule id="100620" level="12">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)bcdedit\s\s\/set\s{default}\srecoveryenabled\sNo</field>
    <description>System recovery disabled. Possible ransomware activity
detected.</description>
    <mitre>
      <id>T1059</id>
    </mitre>
  </rule>

  <rule id="100621" level="12">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)wbadmin\s\sdelete\scatalog\s-quiet</field>
    <description>System catalog deleted. Possible ransomware activity
detected.</description>
    <mitre>
      <id>T1059</id>
    </mitre>
  </rule>
```

```
  <rule id="100622" level="12">

    <if_sid>61603</if_sid>

    <field name="win.eventdata.CommandLine"
type="pcre2">(?i)bcdedit\s\s\/set\s{default}\srecoveryenabled\sNo</field>

    <description>System recovery disabled. Possible ransomware activity
detected.</description>

    <mitre>

      <id>T1059</id>

    </mitre>

  </rule>


  <rule id="100623" level="12">

    <if_sid>92032</if_sid>

    <field name="win.eventdata.CommandLine" type="pcre2">(?i)wevtutil.*cl</field>

    <description>Windows event logs deleted. Possible malicious activity
detected.</description>

    <mitre>

      <id>T1070.001</id>

    </mitre>

  </rule>


<!-- Ransom note file creation -->
```

```
  <rule id="100626" level="10" timeframe="50" frequency="3" ignore="300">

    <if_matched_sid>554</if_matched_sid>

    <same_field>md5</same_field>

    <different_field>file</different_field>

    <description>The file $(file) has been created in multiple directories in a short
time. Possible ransomware activity.</description>

  </rule>


  <rule id="100627" level="7" timeframe="30" frequency="10" ignore="300">

    <if_matched_sid>550</if_matched_sid>

    <field name="file" type="pcre2">(?i)C:\\Users</field>

    <description>Multiple Files modified in the User directory in a short
time.</description>

  </rule>


  <rule id="100629" level="7" timeframe="300" frequency="2" ignore="300">

    <if_matched_sid>63104</if_matched_sid>

    <field name="win.system.message" type="pcre2">(?i)log file was cleared</field>

    <description>Windows Log File Cleared.</description>

    <mitre>

      <id>T1070.001</id>

    </mitre>
```

```
  </rule>

<!-- Detect creation of typical ransom note files -->

  <rule id="100630" level="10">

    <if_group>syscheck</if_group>

    <field name="filename"
type="pcre2">(?i)(README\.txt|HOW_TO_DECRYPT\.txt|RECOVER_FILES\.html)<
/field>

    <description>Possible ransomware ransom note detected:
$(filename)</description>

    <group>ransomware,file_monitoring</group>

    <mitre>

      <id>T1486</id> <!-- Data Encrypted for Impact -->

    </mitre>

  </rule>



  <!-- Detect suspicious ransomware-related file extensions -->

  <rule id="100631" level="10">

    <if_group>syscheck</if_group>

    <field name="filename" type="pcre2">(?i)\.(locked|encrypted|crypt|cry)$</field>

    <description>Suspicious ransomware-related file extension detected:
$(filename)</description>

    <group>ransomware,file_monitoring</group>

    <mitre>

      <id>T1486</id> <!-- Data Encrypted for Impact -->
```

```
    </mitre>
  </rule>

</group>

<group name="ransomware,ransomware_detection">
  <rule id="100628" level="12" timeframe="300" frequency="2" ignore="300">
    <if_matched_group>ransomware_pre_detection</if_matched_group>
<if_sid>100626,100627,100615,100616,100617,100618,100619,100630,100631</if_sid>
    <description>Ransomware activity detected.</description>
  </rule>
</group>
```

```
  GNU nano 8.3                                   /var/ossec/etc/rules/ransomware_rules.xml
<group name="malware,ransomware,ransomware_pre_detection">
<!-- Rules to detect Ransomware attack -->

 <!-- Suspicious command execution -->
 <rule id="100600" level="12">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.parentCommandLine" type="pcre2">(?i)[c-z]:\\\\Windows\\\\System32\\\\svchost\.exe\s-k\sWerSvcGroup</field>
    <field name="win.eventdata.commandLine" type="pcre2">(?i)[c-z]:\\\\Windows\\\\system32\\\\WerFault\.exe\s-pss\s-s\s\d+\s-p\s\d+\s-ip\s\d+</field>
    <description>Possible WerFault DLL Sideloading $(win.eventdata.commandLine).</description>
    <mitre>
      <id>T1546.008</id>
    </mitre>
 </rule>

 <rule id="100601" level="10" >
    <if_sid>61603</if_sid>
    <field name="win.eventdata.parent.image.path" type="pcre2">(?i)regedit.exe</field>
    <field name="win.eventdata.commandLine" type="pcre2">(?i)schtasks.exe \/create.*\\cmd.exe.*start wordpad.exe.*.dll</field>
    <description>Suspicious scheduled task created.</description>
    <mitre>
      <id>T1546.008</id>
    </mitre>
 </rule>

 <rule id="100602" level="7">
   <if_sid>92027</if_sid>
   <field name="win.eventdata.CommandLine" type="pcre2">Install-WindowsFeature.*RSAT-ADPowerShell</field>
   <description>Remote Server Administration Tools installed.</description>
   <mitre>
     <id>T1562</id>
   </mitre>
 </rule>
```

Where:

Rule ID 100600 is triggered when WerFault has been possibly abused via DLL Sideloading.
Rule ID 100601 is triggered when a suspicious scheduled task has been created.
Rule ID 100602 is triggered when malware installs Remote Server Administration Tools.
Rule ID 100603 is triggered when the Windows firewall is disabled.
Rule IDs 100604, 100605, 100606, 100607, 100608, 100609, and 100610 are triggered when Windows Defender service is tampered with.
Rule ID 100611 is triggered when Windows boot configuration data is edited.
Rule ID 100612 is triggered when a new run key is added to the Windows registry.
Rule ID 100613 is triggered when a program is added to Startup programs.
Rule ID 100614 is triggered when Windows WerFault is abused to dump user credentials.
Rule IDs 100615, 100616, 100617, 100618, and 100619 are triggered when there is an attempt to delete Windows volume shadow copy.
Rule ID 100620, 100622 is triggered when BCDEdit changes boot configuration data to inhibit system recovery.
Rule ID 100621 is triggered when wbadmin is used to delete Windows system catalogs.
Rule IDs 100623 and 100629 are triggered when there is an attempt to delete Windows event logs.
Rule ID 100626 is triggered when a file is created in multiple directories within a short period, denoting a possible ransom note creation.
Rule ID 100627 is triggered when multiple files are modified in the user directory within a short period, denoting a possible ransomware file encryption activity.
Rule ID 100628 is triggered when multiple ransomware execution activities are detected, denoting a ransomware activity.
Rule ID 100800 is triggered when the Wazuh Active Response module successfully recovers encrypted files on a monitored endpoint.

Restart the Wazuh manager to apply the configuration changes:

```
systemctl restart wazuh-manager
```

Note: While the detection technique described in this blog post detects multiple variants of Ransomware, it is worth noting that it does not detect all types of Ransomware. This technique was tested with Crosslock, Phobos, BlackCat, Babuk,

AESRTRansomware, AgendaRansomware, BlackBasta, LockBit, and Chaos ransomware variants.

Setting Up the Test Environment

This test is designed to safely generate events on a Windows host that simulate ransomware-like activities. The purpose is to validate Wazuh's detection capabilities, including detection of suspicious volume shadow copy operations, BCDEdit modifications, and other potential ransomware behaviors. All actions are simulated and **will not harm the system**.

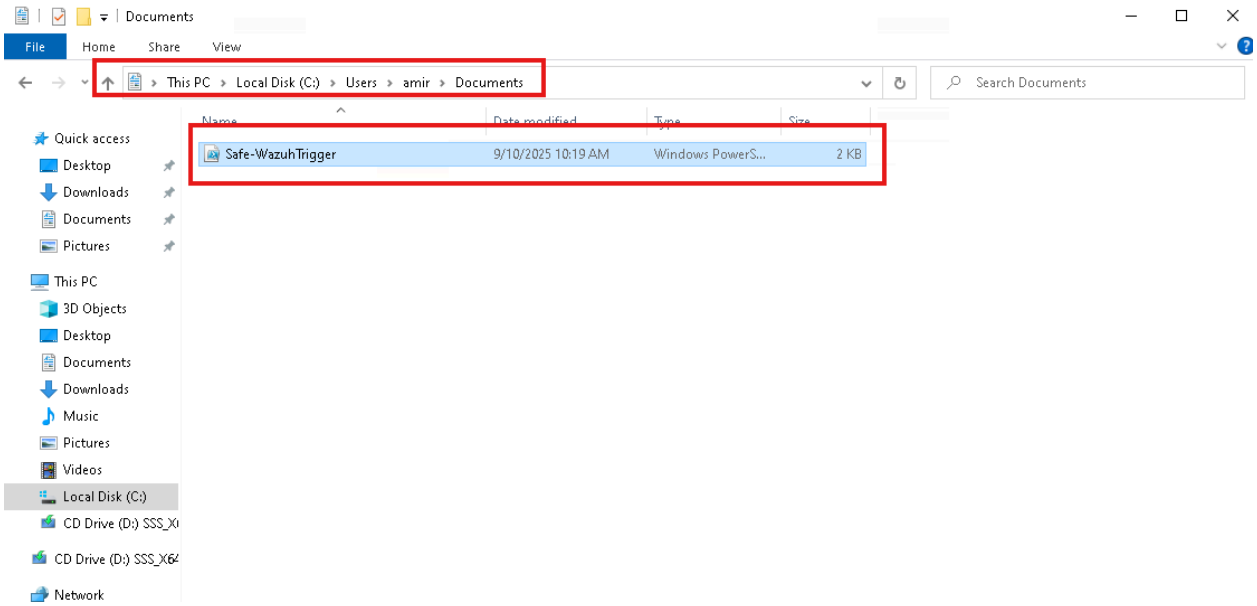
Script is following:

```
# Safe-WazuhTrigger.ps1

# Ensure Administrator

if (-not ([Security.Principal.WindowsPrincipal]
[Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([Security.Principal.Wind
owsBuiltinRole] "Administrator")) {

    Write-Host "Run PowerShell as Administrator!" -ForegroundColor Red

    exit

}

Write-Host "Triggering Wazuh rules 100616-100620 safely..."

Start-Process -FilePath "cmd.exe" -ArgumentList "/c echo vssadmin delete shadows
/all /quiet" -NoNewWindow

Write-Host "100616 triggered"


Start-Process -FilePath "cmd.exe" -ArgumentList "/c echo Shadowcopy Delete"
-NoNewWindow

Write-Host "100617 triggered"
```

```
Start-Process -FilePath "cmd.exe" -ArgumentList "/c echo wmic shadowcopy delete" -NoNewWindow

Write-Host "100618 triggered"


Start-Process -FilePath "cmd.exe" -ArgumentList "/c echo delete shadows" -NoNewWindow

Write-Host "100619 triggered"

Start-Process -FilePath "cmd.exe" -ArgumentList "/c echo bcdedit /set {default} recoveryenabled No" -NoNewWindow

Write-Host "100620 triggered"

Write-Host "All test events safely triggered. Check Wazuh dashboard."
```

Execution Instructions

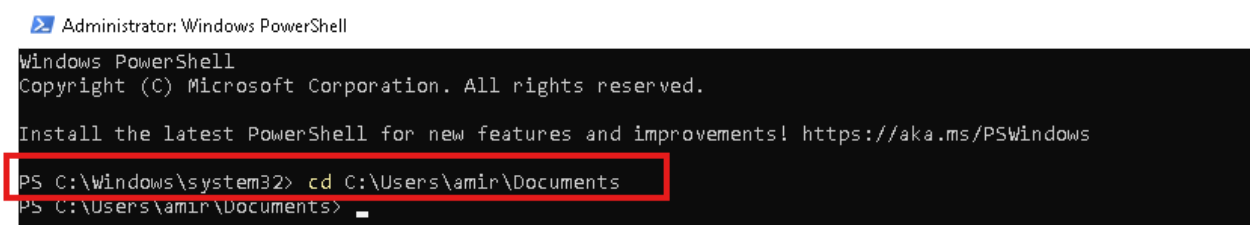
Save the script as:
C:\Users\amir\Documents\Safe-WazuhTrigger.ps1



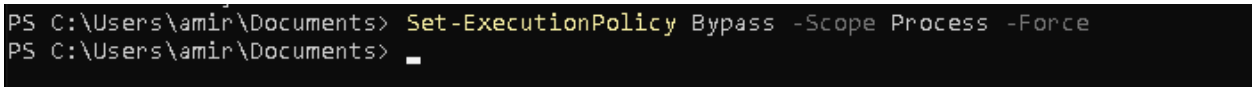
Open PowerShell as Administrator.

Navigate to the script directory:

cd C:\Users\amir\Documents



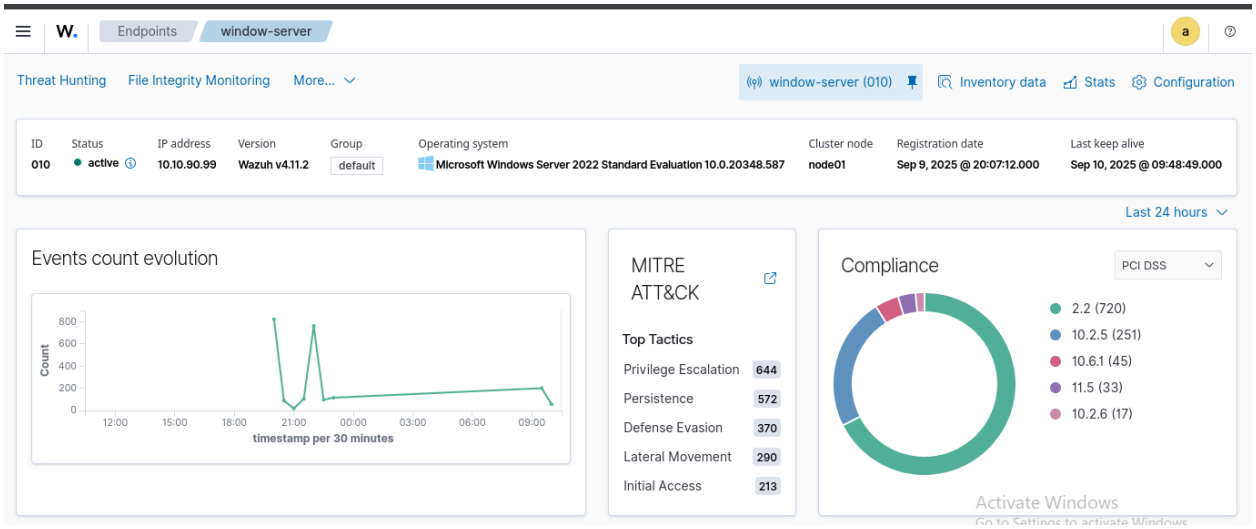
Temporarily bypass the execution policy:
Set-ExecutionPolicy Bypass -Scope Process -Force

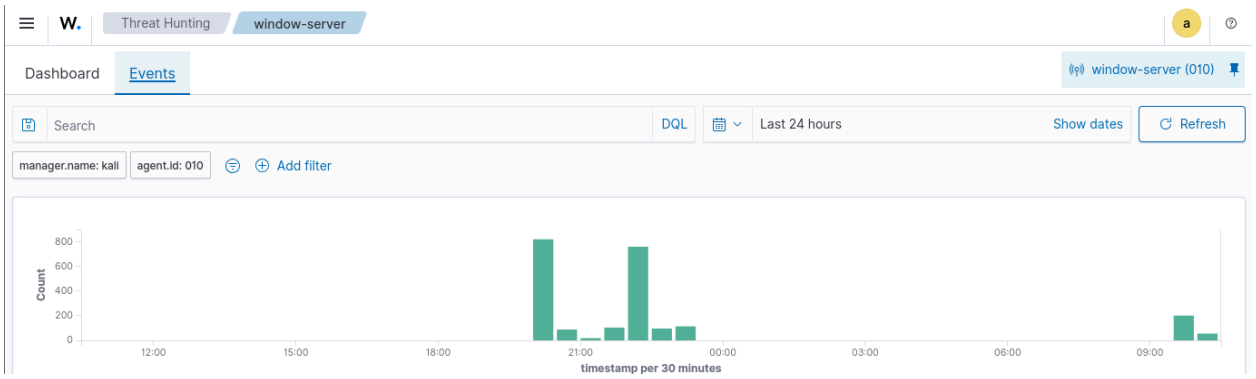


Run the script:
.\Safe-WazuhTrigger.ps1

```
Administrator: Windows PowerShell
PS C:\Users\amir\Documents> .\Safe-WazuhTrigger.ps1
Triggering Wazuh rules 100616-100620 safely...
100616 triggered
100617 triggered
100618 triggeredShadowcopy Delete

vssadmin delete shadows /all /quiet
wmic shadowcopy delete
100619 triggered
100620 triggered
delete shadows
All test events (100616-100620) safely triggered. Check Wazuh dashboard.
PS C:\Users\amir\Documents> bcdedit /set {default} recoveryenabled No
```

After execution, verify that simulated ransomware events appear in the Wazuh dashboard.





Threat Hunting | window-server

2,247 hits

Sep 9, 2025 @ 10:23:05.359 - Sep 10, 2025 @ 10:23:05.359

Export Formatted | 1116 available fields | Columns | Density | **1 fields sorted** | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Sep 10, 2025 @ 10:22:40.9...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:22:40.9...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:22:40.8...	window-server	Volume shadow copy deleted using Cmd.Exe. Potential ransomware activity detected.	12	100617
Sep 10, 2025 @ 10:22:40.7...	window-server	Volume shadow copy deleted using Cmd.Exe. Potential ransomware activity detected.	12	100617
Sep 10, 2025 @ 10:22:40.7...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:22:29.9...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:22:29.9...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:22:29.8...	window-server	Ransomware activity detected.	12	100628
Sep 10, 2025 @ 10:22:29.8...	window-server	Volume shadow copy deleted using Cmd.Exe. Potential ransomware activity detected.	12	100617
Sep 10, 2025 @ 10:22:29.8...	window-server	Powershell process spawned Windows command shell instance	4	92004
Sep 10, 2025 @ 10:20:48.7...	window-server	Executable file dropped in folder commonly used by malware	15	92213

Discover | wazuh-alerts-4.x-2025.09.10#s74TMpkBtKiptD09h--1

Table | JSON

@timestamp	Sep 10, 2025 @ 10:22:40.825
_index	wazuh-alerts-4.x-2025.09.10
agent.id	010
agent.ip	10.10.90.99
agent.name	window-server
data.win.eventdata.commandLine	\"C:\\Windows\\system32\\cmd.exe\" /c echo wmic shadowcopy delete
data.win.eventdata.company	Microsoft Corporation
data.win.eventdata.currentDirectory	C:\\Users\\amir\\Documents\\
data.win.eventdata.description	Windows Command Processor
data.win.eventdata.fileVersion	10.0.20348.1 (WinBuild.160101.0800)
data.win.eventdata.hashes	SHA1=2ED89B5430C775306B316BA3A926D7DE4FE39FC7,MD5=E7A6B1F51EFB405287A8048CFA4690F4,SHA256=EB71EA69DD19F728AB9240565E8C7EFB59821E19E3788E289301E1E74940C208,IMPHASH=D60B77062898DC6BFAE7FE11A0F8806C
data.win.eventdata.image	C:\\Windows\\System32\\cmd.exe
data.win.eventdata.integrityLevel	High
data.win.eventdata.ruleName	technique_id=T1490,technique_name=Inhibit System Recovery
data.win.eventdata.terminalSessionId	1
data.win.eventdata.user	AGENT\\amir
data.win.eventdata.utcTime	2025-09-10 05:22:39.642
data.win.system.channel	Microsoft-Windows-Sysmon/Operational
data.win.system.computer	agent
data.win.system.eventID	1
data.win.system.eventRecordID	10526
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"Process Create: RuleName: technique_id=T1490,technique_name=Inhibit System Recovery UtcTime: 2025-09-10 05:22:39.642 ProcessGuid: {ebb0310a-0b1f-68c1-8701-000000000800} ProcessId: 3008 Image: C:\Windows\System32\cmd.exe FileVersion: 10.0.20348.1 (WinBuild.160101.0800)
data.win.system.opcode	0
data.win.system.processID	2080

id	1757481749.1745767
input.type	log
location	EventChannel
manager.name	kali
rule.description	Volume shadow copy deleted using Cmd.Exe. Potential ransomware activity detected.
rule.firedtimes	2
rule.groups	malware, ransomware, ransomware_pre_detection
rule.id	100617
rule.level	12
rule.mail	true
rule.mitre.id	T1490, T1059.003
rule.mitre.tactic	Impact, Execution
rule.mitre.technique	Inhibit System Recovery, Windows Command Shell
timestamp	Sep 10, 2025 @ 10:22:29.855

Summary:

This project is designed to provide comprehensive ransomware detection and mitigation on Windows endpoints using Wazuh, offering proactive protection against malicious software that encrypts, locks, or damages critical files. To ensure robust monitoring, the system leverages File Integrity Monitoring (FIM) to track changes in user directories, system files, and application folders, combined with log analysis of Windows Security, System, and Application events for suspicious activities such as privilege escalation, deletion of shadow copies, and disabling of security tools. The system integrates with VirusTotal to validate detected files and uses Wazuh Active Response to automatically quarantine or remove threats, minimizing potential data loss.

All configurations are applied on a dedicated Wazuh agent hosted on Windows Server, with the manager deployed on Kali Linux, maintaining a clear separation between monitoring and response components. Custom rules are created to detect ransomware behaviors, including unusual file renaming, mass file modifications, ransom note creation, and suspicious execution patterns, providing early alerts and actionable intelligence.

A safe testing environment with simulated ransomware events ensures that detection rules and response mechanisms are validated without risking system integrity. All monitoring and active response activities are logged for auditing and continuous improvement, enabling administrators to track incidents, refine detection logic, and respond swiftly to emerging ransomware threats.

This setup follows best practices for dependency management, configuration, and system hardening, allowing scalability, automation, and adaptability to new ransomware variants, strengthening overall endpoint security and reducing operational risk.